

Data Analytics with Power BI

Project Report

(Project Semester: July-Dec 2025)

Indian Crime Data Analytics

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CERTIFICATE

This is to certify that **Sudhir Singh** bearing Registration no. **12318709** has completed **INT374** project titled, “**Indian Crime Data Analytics**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature of the Supervisor

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DECLARATION

I, Sudhir Singh, student of Bachelor of Technology – B. Tech under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.



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Signature

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Sudhir Singh

ACKNOWLEDGEMENT

It gives me immense pleasure to present this project report on “**Crime Data Analytics Dashboard using Microsoft Power BI**”, undertaken as a part of my academic curriculum at **Lovely Professional University**. This project has been a highly enriching learning experience, enabling me to gain hands-on exposure to real-world data analytics by working with large crime datasets and transforming raw data into meaningful insights.

Through this project, I explored practical applications of **data cleaning, data modeling, DAX functions, data visualization, and interactive dashboard development** using Microsoft Power BI. The project enhanced my ability to analyze crime trends, identify patterns, and present data-driven insights in a visually intuitive and interactive manner to support informed decision-making.

I would like to express my sincere gratitude to **Ms. Baljinder Kaur**, my respected professor and project mentor, for her continuous guidance, valuable suggestions, and unwavering support throughout the development of this project. Her insightful feedback, domain expertise, and constructive advice played a pivotal role in shaping the direction of this work and motivated me to improve both my technical and analytical skills.

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1. Introduction

1.1 Background and Motivation

With the rapid growth of digital records, crime-related data has become large, complex, and multidimensional. In India, crime statistics are recorded across various parameters such as **crime type, state, district, year, age group, and gender**, resulting in datasets that are difficult to analyze using traditional static tools like spreadsheets or manual reports.

Analyzing such datasets requires not only visualization but also **dynamic calculations, aggregations, and trend analysis**. Business Intelligence tools address this challenge by combining data modeling, calculations, and interactive reporting. The motivation behind this project is to apply **Microsoft Power BI** as a comprehensive analytics platform to convert raw crime data into meaningful insights using **Data Analysis Expressions (DAX)** and interactive dashboards.

Power BI allows the creation of calculated measures such as **total crimes, crime growth percentage, age-wise victim counts, and gender-wise distributions**, which are not directly available in raw datasets. These calculations are performed using DAX, enabling deeper and more flexible analysis. This project focuses on analyzing **Indian crime data from 2001 to 2013** and presenting insights in an intuitive, interactive, and analytical format.

1.2 Problem Statement

Although large volumes of crime data are available, extracting accurate and actionable insights remains challenging due to the absence of:

- Dynamic calculations across multiple tables
- Time-based trend analysis
- Interactive filtering and drill-down capabilities

Without the use of analytical expressions and a structured data model, it becomes difficult to compute metrics such as **total crimes across years, comparative crime growth, age-wise victim trends, and gender-wise crime distribution**. Therefore, there is a need for an analytical system that integrates **data modeling with DAX-based measures** to support efficient crime data analysis.

1.3 Objectives of the Project

The objectives of this project are:

- To analyze Indian crime data using Microsoft Power BI
 - To design an interactive, multi-page dashboard
 - To study crime trends over different years
 - To create DAX measures for calculating total crimes, victims, and growth rates
 - To analyze crime trends across years using time-based calculations
 - To study crime distribution by state, district, age group, and gender
 - To apply data modeling concepts using fact and dimension tables
-

1.4 Scope of the Project

The scope of this project includes **descriptive and diagnostic crime data analysis** using historical data from **2001 to 2013**. The project emphasizes:

- Data modeling and relationships
- DAX-based calculated measures
- Interactive visual analysis

Advanced predictive modeling and real-time analytics are outside the scope of this project.

1.5 Significance of DAX in the Project

Data Analysis Expressions (DAX) plays a crucial role in this project by enabling the creation of:

- Aggregated metrics such as total crimes and total victims
- Percentage-based measures such as crime growth rate
- Dynamic calculations that respond to slicers and filters

DAX allows the dashboard to provide **real-time recalculation of values** based on user interaction, making the analysis more accurate and insightful compared to static reporting methods.

2. LITERATURE REVIEW

2.1 Crime Data Analytics and Monitoring Systems

Crime data analytics has emerged as an essential domain for understanding criminal behavior, identifying crime-prone regions, and supporting policy formulation. Various studies highlight that systematic analysis of crime data helps law enforcement agencies detect trends, allocate resources efficiently, and take preventive measures.

Traditional crime monitoring systems relied heavily on static reports and manual record analysis, which limited the ability to perform comparative and trend-based analysis. With the increase in data volume and complexity, modern crime monitoring systems increasingly use **data analytics dashboards** to visualize crime statistics across geographical, temporal, and demographic dimensions.

Research indicates that interactive dashboards improve situational awareness by allowing users to explore crime data dynamically rather than relying on fixed summaries. Such systems are widely adopted in public safety and governance domains for effective crime analysis.

2.2 Role of Data Analytics and Business Intelligence in Crime Analysis

Data analytics plays a vital role in converting raw crime records into meaningful insights. Through techniques such as **data cleaning, aggregation, filtering, and trend analysis**, analysts can uncover hidden patterns in crime datasets. Business Intelligence (BI) tools enhance this process by integrating analytics with visualization and interactivity.

Several studies emphasize that BI platforms enable:

- Multi-dimensional analysis of crime data
- Comparison across time periods, locations, and categories
- Faster interpretation of complex datasets

By using calculated metrics and dynamic filters, BI tools allow stakeholders to evaluate crime growth, identify dominant crime types, and understand demographic impact. This makes BI-driven crime analysis more effective than traditional spreadsheet-based approaches.

2.3 Microsoft Power BI as a Crime Analytics Tool

Microsoft Power BI is widely recognized as a powerful Business Intelligence tool used for data modeling, visualization, and reporting. It supports integration of data from multiple sources and provides features such as **Power Query, DAX, interactive visuals, and drill-down analysis**, making it suitable for crime data analytics.

In crime analysis projects, Power BI enables:

- Creation of structured data models using fact and dimension tables
- Development of DAX measures for calculating totals, percentages, and growth rates
- Design of interactive dashboards with slicers, filters, and maps

The ability of Power BI to recalculate DAX measures dynamically based on user selection allows analysts to explore crime data across different perspectives such as **state-wise, district-wise, year-wise, age-wise, and gender-wise**. Hence, Power BI is an effective platform for large-scale crime data analysis and visualization.

2.4 Importance of DAX in Analytical Reporting

Data Analysis Expressions (DAX) is a formula language used in Power BI to create calculated measures and columns. Literature highlights that DAX plays a critical role in analytical reporting by enabling advanced calculations that are not directly available in raw datasets.

In the context of crime analytics, DAX is used to:

- Calculate total crimes and total victims
- Measure year-wise crime growth and decline
- Perform context-aware calculations based on filters and slicers

DAX enhances analytical depth by allowing measures to respond dynamically to user interaction, which significantly improves the accuracy and usefulness of dashboards.

2.5 Summary of Literature Review

From the literature reviewed, it is evident that:

- Crime data analytics is essential for understanding crime trends and patterns
- Business Intelligence tools significantly improve crime analysis efficiency
- Power BI, combined with DAX, provides a robust platform for interactive crime analytics
- Analytical dashboards offer superior insights compared to traditional reporting methods

These findings support the selection of **Microsoft Power BI and DAX** as the core tools for this project and justify their application in analyzing Indian crime data.

Some used DAX Functions Snapshots

```
1 Total Victims =
2 SUM(fact_murdervictim[TOTAL])
3
```

```
1 Total Police Complaints =
2 SUM(Fact_PoliceComplaints[CPA_-_Cases_Registered])
3
```

```
1 Crime Growth % =
2 VAR CurrentYear = MAX(dim_year[Year])
3 VAR CurrentValue = [Total Crime]
4 VAR PreviousValue =
5     CALCULATE(
6         [Total Crime],
7         FILTER(
8             ALL(dim_year),
9             dim_year[Year] = CurrentYear - 1
10        )
11    )
12 RETURN
13 IF(
14     ISBLANK(PreviousValue),
15     0,
16     DIVIDE(CurrentValue - PreviousValue, PreviousValue)
17 )
18
```

3. SOURCE OF DATASET

3.1 Dataset Source

The dataset used in this project has been obtained from **open-source crime data repositories** that provide access to publicly available Indian crime statistics for research and academic purposes. The data is based on crime records compiled from official sources such as the **National Crime Records Bureau (NCRB), Government of India**, and made available in a cleaned and structured format.

For this project, the dataset was downloaded from **NCRB**, a widely used data science platform that hosts government and research datasets for analytical use.

Dataset Source Link:

<https://www.data.gov.in/ministrydepartment/national-crime-records-bureau-ncrb>

This dataset contains detailed crime records across Indian states and districts over multiple years, making it suitable for crime trend analysis and dashboard-based visualization using Power BI.

3.2 Dataset Description

The dataset covers crime statistics for the period **2001 to 2013** and includes information related to:

- Year of crime occurrence
- State and district
- Type of crime
- Gender of victims
- Age group of victims

To enable efficient analysis, the data has been organized into multiple **fact tables** representing different crime categories (such as crimes against women, juvenile crimes, fraud, kidnapping, murder, and custodial deaths) along with supporting **dimension tables** including **Year, State, District, Crime Type, Gender, and Age Group**.

3.3 Data Collection Methodology

The original crime data was collected by government authorities through official reporting mechanisms and later published as annual crime statistics. These records were aggregated and released in tabular format, which were then curated and published on Kaggle for public access.

The dataset was downloaded in CSV/Excel format and imported into **Microsoft Power BI Desktop** for further analysis.

3.4 Data Preprocessing and Transformation

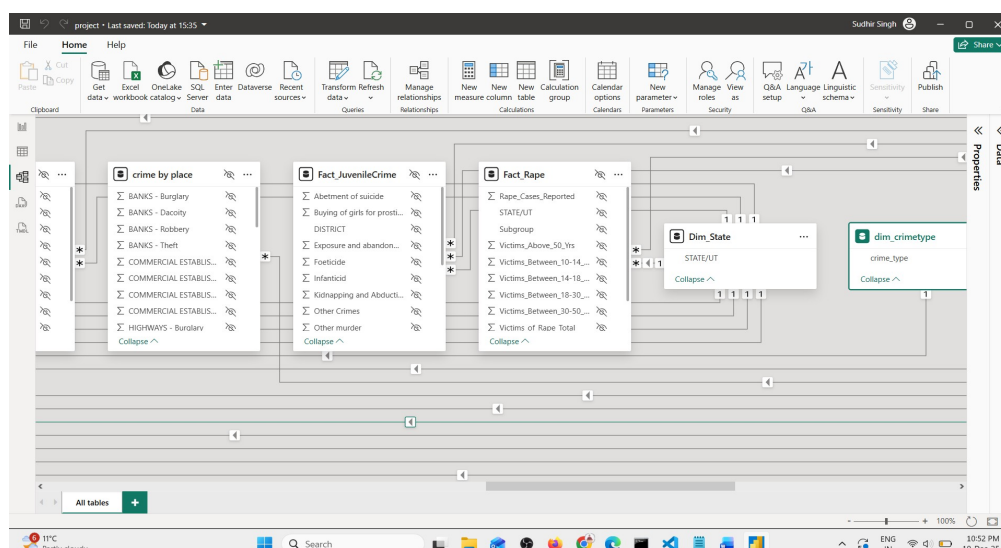
Using **Power Query Editor**, the following preprocessing steps were performed:

- Removal of duplicate and incomplete records
- Handling of missing and null values
- Standardization of column names
- Conversion of data types
- Creation of derived columns for total crime calculations

These steps ensured the data was clean, consistent, and ready for analytical modeling.

3.5 Data Modeling and Integration

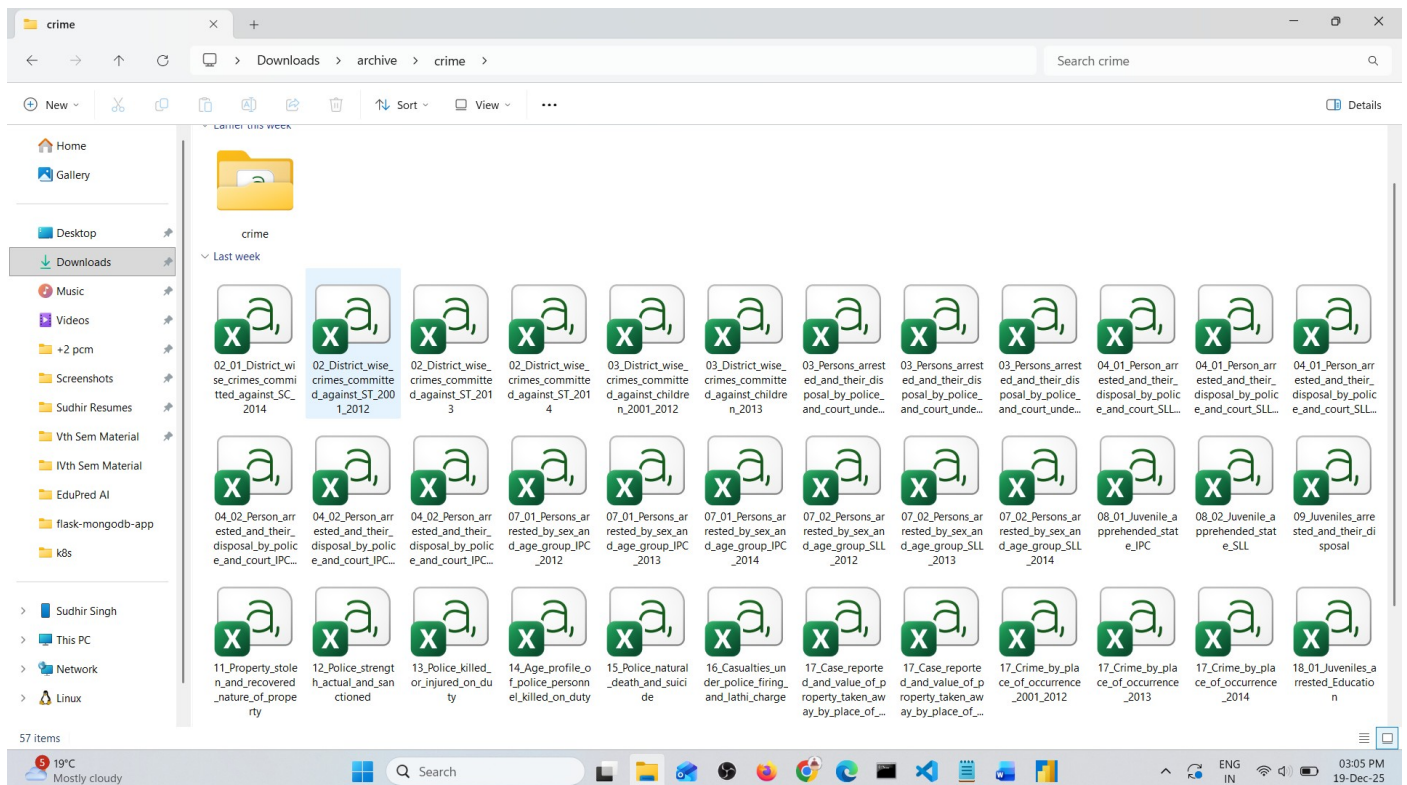
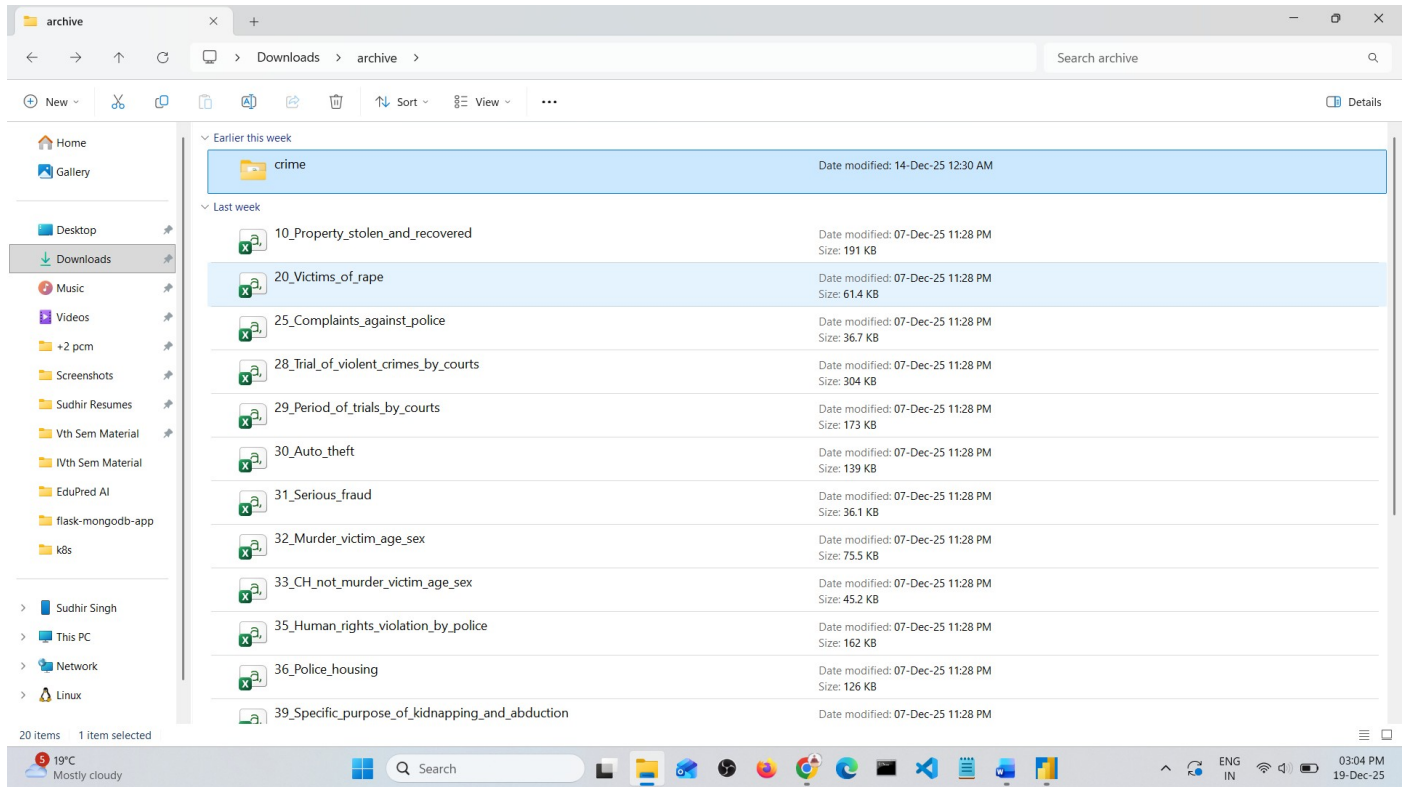
The dataset was structured using a **fact–dimension data model**, where multiple fact tables were connected to shared dimension tables through one-to-many relationships. This modeling approach supports accurate aggregation and **DAX-based dynamic calculations** across visuals.



Data modelling

3.6 Summary

In summary, the crime dataset used in this project was sourced from **Kaggle (NCRB-based Indian crime statistics)**. Proper preprocessing, data modeling, and integration were carried out to prepare the dataset for interactive crime data analytics using Microsoft Power BI.



4. DATA PREPROCESSING & EDA

4.1 Data Preprocessing

Data preprocessing is a crucial step in crime data analytics, as raw crime datasets often contain inconsistencies, missing values, and unstructured fields. Before performing any analysis or visualization, the dataset was thoroughly cleaned and prepared using **Power Query Editor in Microsoft Power BI**.

4.1.1 Data Cleaning

The following data cleaning steps were performed:

- Removal of duplicate records to avoid double counting of crime cases
- Identification and handling of missing or null values
- Elimination of irrelevant columns not required for analysis
- Correction of inconsistent naming conventions for states, districts, and crime types

These steps ensured data accuracy and reliability.

4.1.2 Data Transformation

Data transformation was carried out to make the dataset suitable for analytical modeling:

- Standardization of column names across multiple datasets
- Conversion of data types (e.g., year to numeric, crime counts to integers)
- Splitting and merging columns where required
- Filtering records to retain only valid years (2001–2013)

Power Query's transformation features ensured that all datasets followed a consistent structure.

4.1.3 Data Integration

Multiple crime-related datasets were imported and integrated within Power BI. These datasets were merged logically by linking them to common **dimension tables** such as:

- Year
- State

- District
- Crime Type
- Gender
- Age Group

This integration enabled unified analysis across different crime categories.

4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and patterns within the crime data before drawing conclusions. Power BI visuals and summary measures were extensively used for this purpose.

4.2.1 Understanding Data Distribution

Initial analysis focused on:

- Total crime counts across years
- Distribution of crimes across states and districts
- Crime-wise contribution to overall crime volume

Simple bar charts, tables, and KPI cards were used to observe the spread and magnitude of crime incidents.

4.2.2 Time-Based Analysis

Year-wise analysis revealed:

- Overall growth in crime trends over time
- Fluctuations in specific crime categories
- Periods with higher or lower crime occurrences

Line charts were used to visualize long-term crime trends from 2001 to 2013.

4.2.3 Demographic Analysis

EDA also focused on demographic aspects:

- **Gender-wise crime distribution** to understand victim patterns
- **Age-wise analysis** to identify vulnerable age groups

Pie charts and bar charts helped in comparing demographic contributions to total crimes.

4.2.4 Geographic Analysis

Geographical analysis was conducted using:

- State-wise and district-wise crime comparisons
- Identification of high-crime regions

Map visuals and region-wise bar charts highlighted crime concentration areas across India.

4.3 Feature Engineering and DAX Measures

To support advanced analysis, additional features and measures were created using **DAX (Data Analysis Expressions)**.

Key measures include:

- **Total Crimes** – aggregated crime counts across all categories
- **Total Victims** – sum of reported victims
- **Crime Growth Rate (%)** – year-on-year change in crime occurrences

These DAX measures enable dynamic recalculation of values based on slicers and filters, enhancing analytical flexibility.

4.4 Key Insights from EDA

The exploratory analysis revealed several important insights:

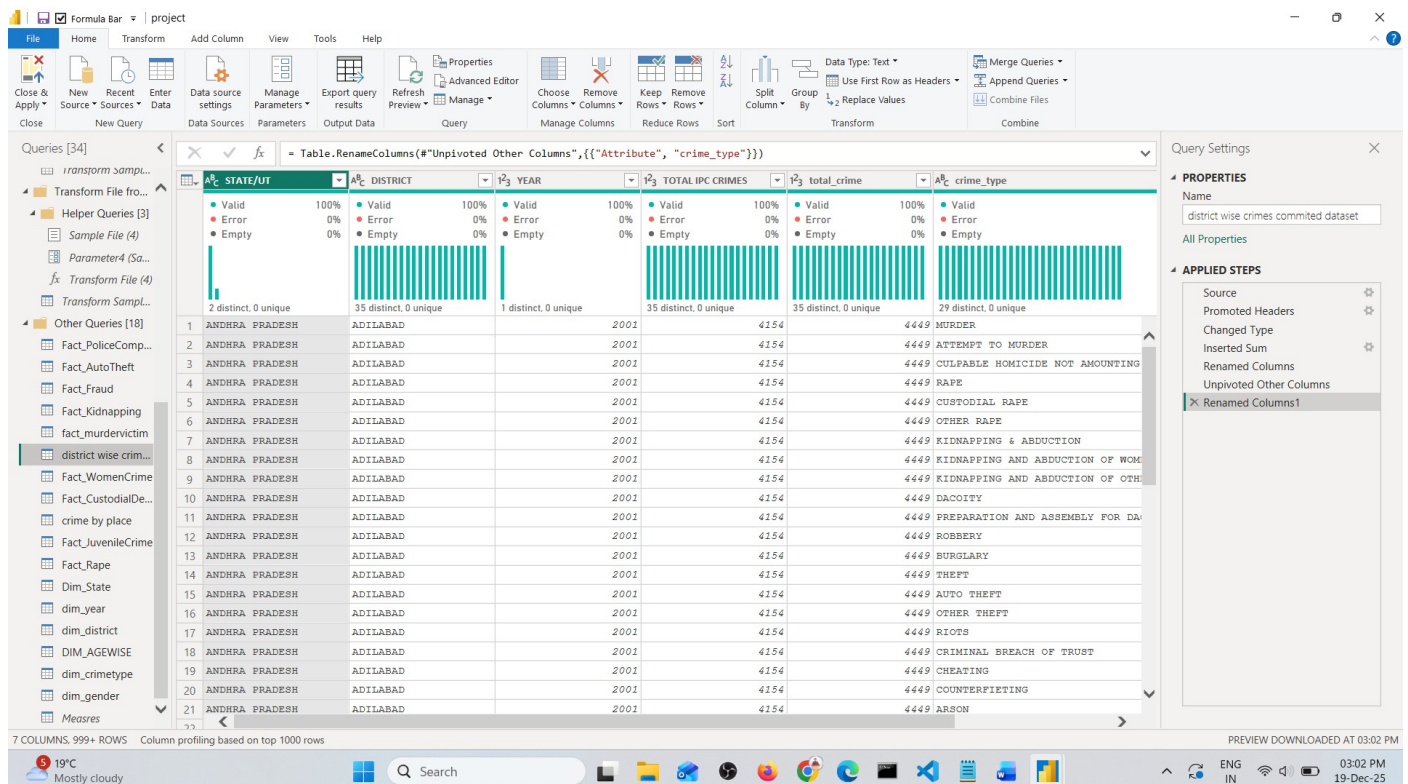
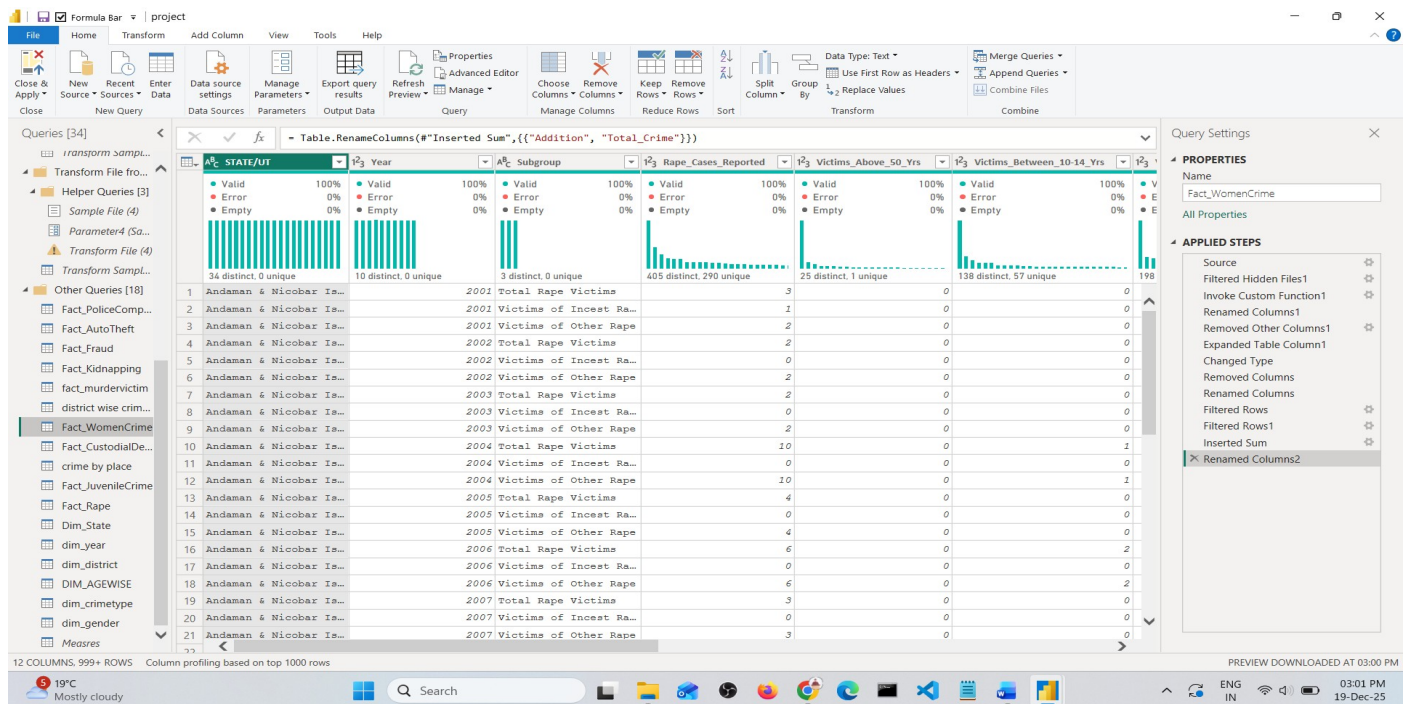
- Crime rates show a generally increasing trend over the years
- Certain states consistently report higher crime volumes
- Specific crime categories dominate overall crime statistics
- Age group and gender play a significant role in crime distribution

These insights guided the design of interactive dashboards and informed deeper analysis in subsequent chapters.

4.5 Summary

In this chapter, raw crime data was cleaned, transformed, and explored using Power BI. Data preprocessing ensured accuracy and consistency, while EDA provided a foundational understanding of crime trends, distributions, and patterns. The insights obtained from this stage formed the basis for detailed analysis and visualization presented in the dashboard.

Snapshot:



5. ANALYSIS ON DATASET

5.1 Introduction to Analysis Objectives

The main objective of this chapter is to analyze crime data using **interactive charts and visualizations** created in Microsoft Power BI. Visual analytics helps in identifying trends, patterns, and comparisons that are difficult to observe through raw data tables.

The analysis focuses on understanding:

- Overall crime magnitude
- Crime trends over time
- Geographic distribution of crimes
- Demographic impact based on gender and age
- Interactive filtering using slicers

5.2 Dashboard Overview and Visual Layout

The Power BI dashboard is designed using a structured layout consisting of **KPI cards, charts, maps, and slicers**. Each visual plays a specific role in presenting crime-related insights clearly and effectively.

SCREENSHOT



5.3 KPI Cards (DAX-Based Summary Visuals)

Visualization Used: KPI / Card Visuals

KPI cards are used to display **high-level summary metrics** calculated using DAX measures. These visuals provide instant insights without requiring deep exploration.

The following KPI cards are included:

- **Total Crimes** – displays the total number of reported crimes
- **Total Victims** – shows the total number of victims
- **Crime Growth (%)** – represents the year-on-year growth in crime cases

These cards dynamically update based on slicers, making them highly interactive.

5.4 Year-wise Crime Trend Analysis

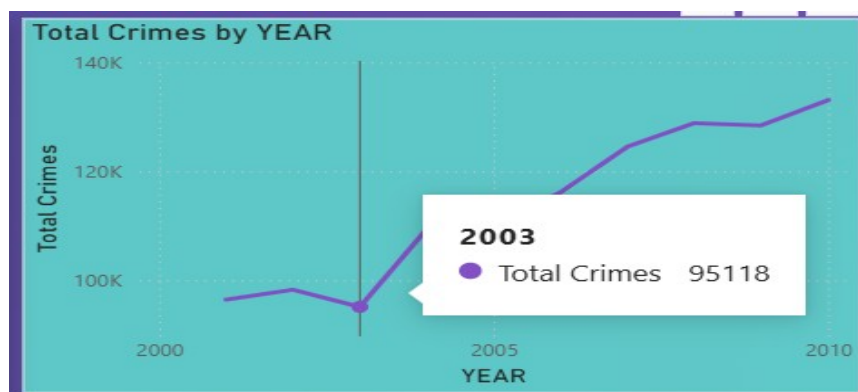
Visualization Used: Line Chart

A **line chart** is used to analyze crime trends over time because it effectively represents changes across continuous time periods.

This visualization shows:

- Growth or decline in crime cases from 2001 to 2013
- Year-wise comparison of total crimes
- Identification of peak and low crime periods

The line chart clearly highlights long-term trends and fluctuations in crime data.



SCREENSHOT

5.5 State-wise and District-wise Crime Distribution

Visualizations Used: Bar Chart and Map Visual

To compare crime distribution across regions:

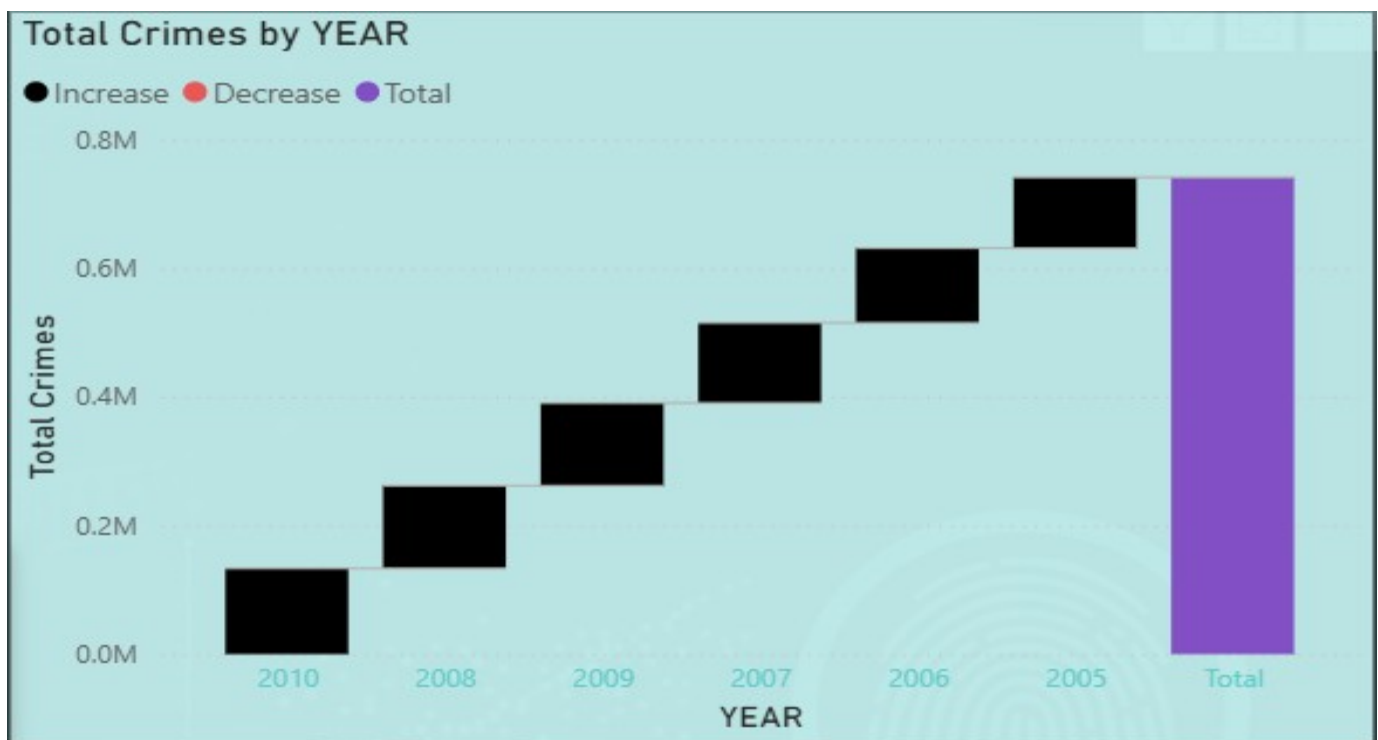
- **Bar charts** are used for state-wise and district-wise comparison
- **Map visuals** are used to represent geographic crime intensity

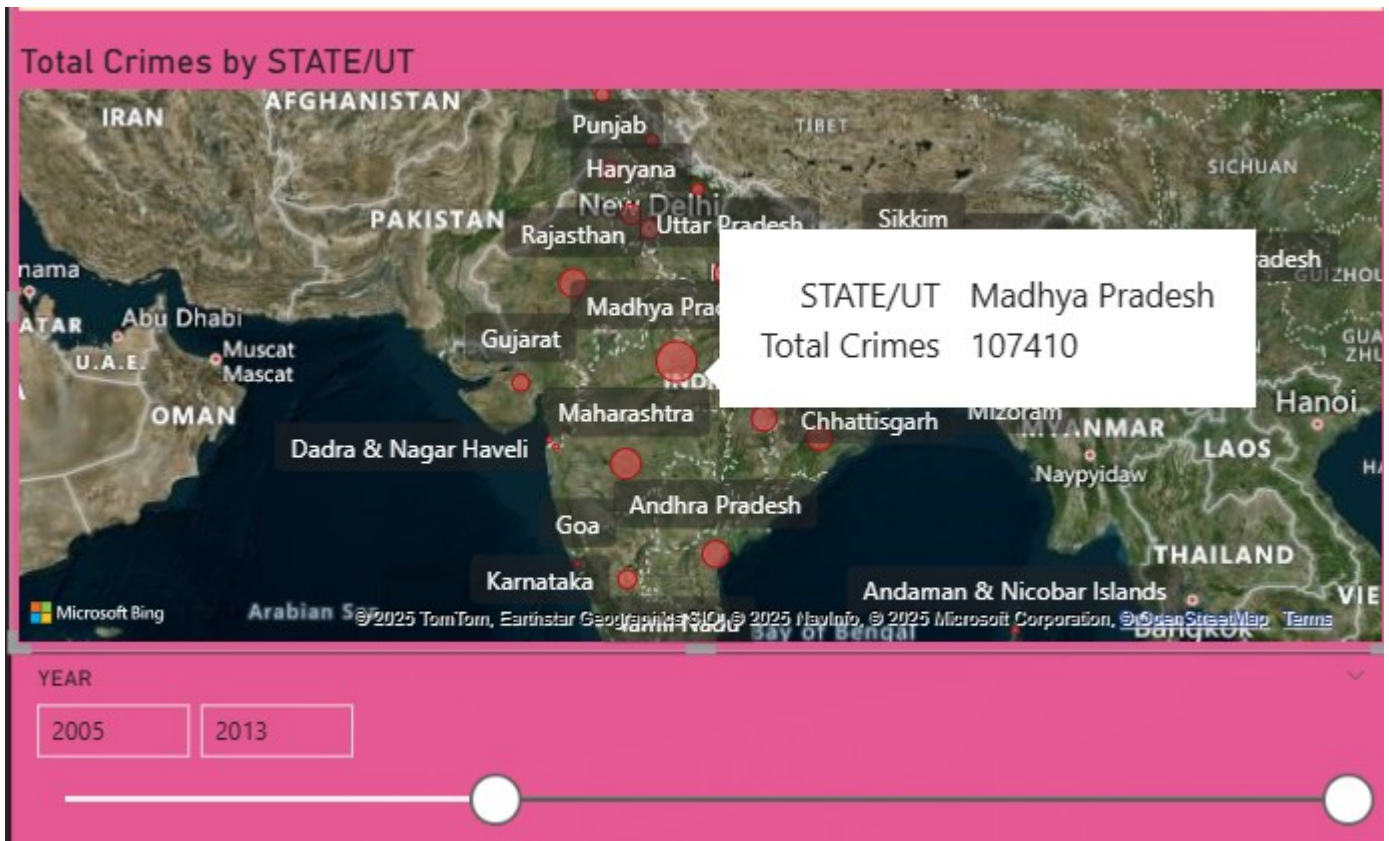
Bar charts help in ranking states and districts based on crime count, while maps provide a spatial understanding of crime concentration.

Key insights include:

- Identification of high-crime states
- Regional imbalance in crime distribution
-

SCREENSHOT





5.6 Crime Type Analysis

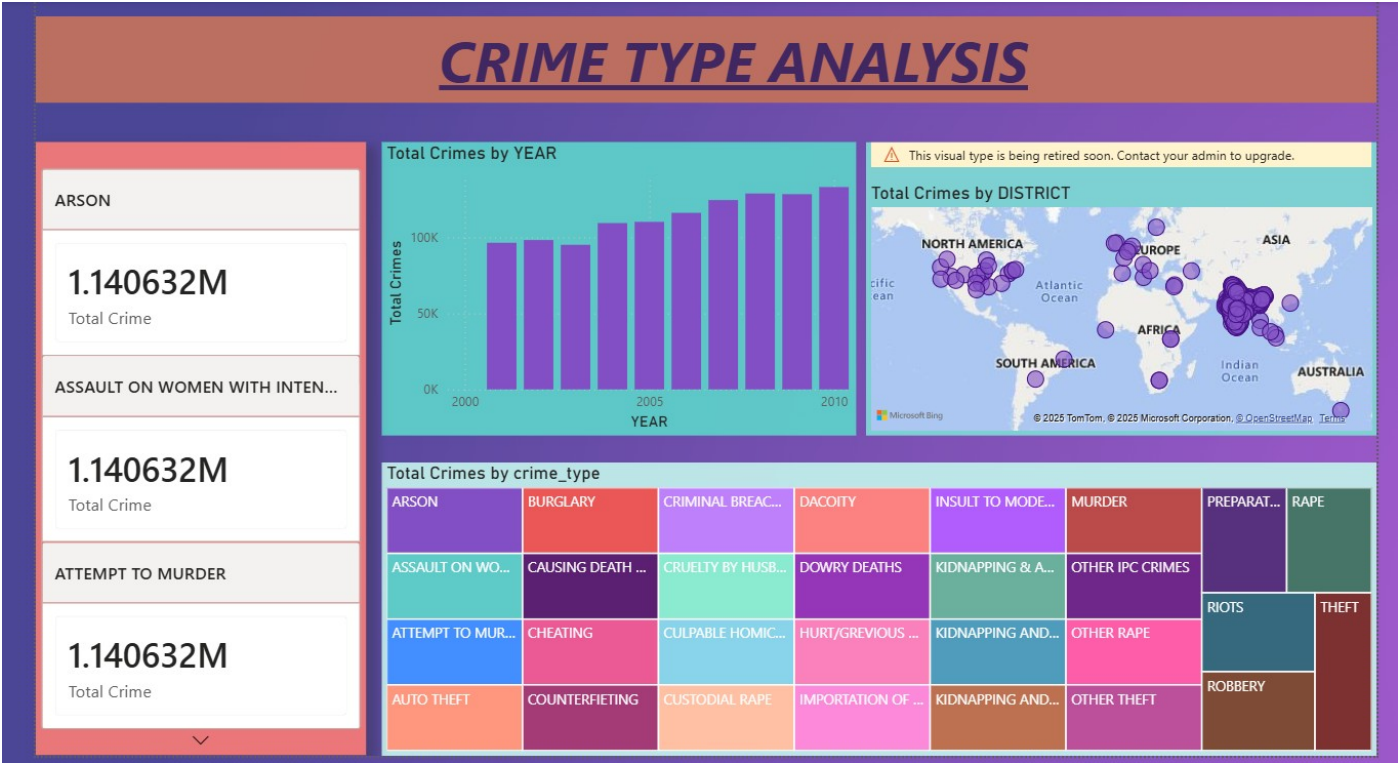
Visualization Used: Bar Chart / Column Chart

Crime type analysis is performed using **bar or column charts** to compare different crime categories such as:

- Crimes against women
- Juvenile crimes
- Fraud
- Kidnapping
- Murder and custodial deaths

This visualization helps in identifying **dominant crime categories** contributing most to total crime volume.,

SCREENSHOT



5.7 Gender-wise and Age-wise Crime Analysis

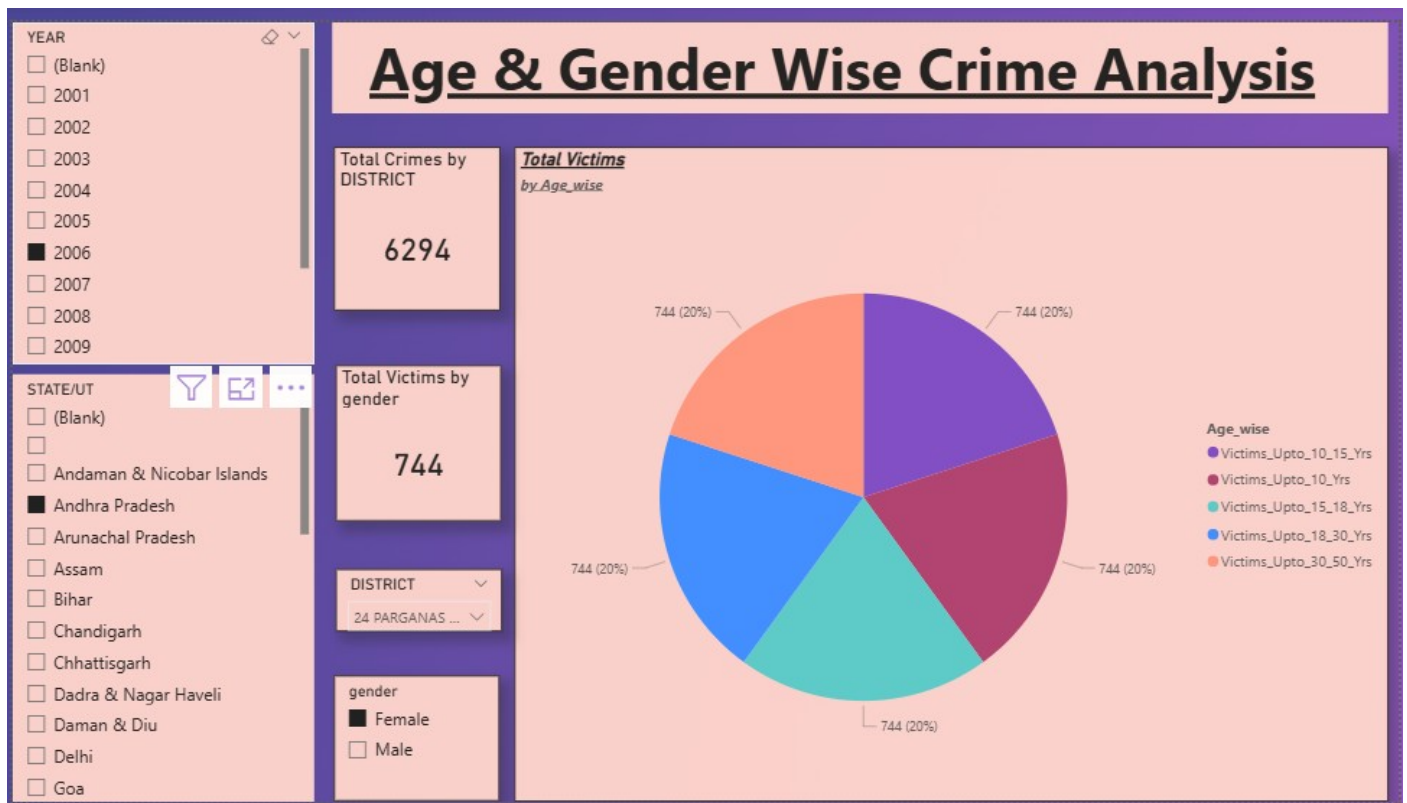
Visualizations Used: Pie Chart and Bar Chart

Demographic analysis is carried out using:

- **Pie charts** for gender-wise crime distribution
- **Bar charts** for age-wise victim analysis

Pie charts clearly represent proportional differences between genders, while bar charts allow comparison across age groups to identify vulnerable populations.

SCREENSHOT



5.8 Interactive Filters and Slicers

Visualization Used: Slicers

Slicers are used to make the dashboard **interactive and dynamic**. The following slicers are included:

- Year
- State
- District
- Crime Type
- Gender
- Age Group

Applying slicers updates all charts and KPIs simultaneously, enabling focused and drill-down analysis.

5.9 Summary of Visual Analysis

This chapter demonstrated how different **charts and visualizations** were used to analyze crime data effectively. By combining **KPI cards, line charts, bar charts, pie charts, maps, and slicers**, the dashboard provides a comprehensive and interactive crime analytics solution.

The visual insights obtained in this chapter form the basis for discussion and interpretation in the next chapter.

-

6. Discussion and Interpretation

6.1 Overview of Analysis Outcomes

This chapter focuses on interpreting the results obtained from the crime data analysis and Power BI visualizations developed in the previous chapters. The purpose of this discussion is to translate raw numbers, charts, and dashboards into meaningful insights that help understand crime patterns, trends, and relationships. The interpretations are based on interactive visualizations such as bar charts, line charts, maps, slicers, and KPI cards created in Power BI.

6.2 Crime Trends Over Time

The time-based analysis of crime data reveals clear variations in crime occurrences across different years and months.

- The line chart representing **crime count over years** shows that crime rates do not remain constant and tend to fluctuate over time.
- Certain years show noticeable peaks, indicating periods of increased criminal activity. These spikes may be influenced by factors such as population growth, economic conditions, or changes in law enforcement reporting.
- Monthly analysis highlights **seasonal trends**, where specific months record higher crime rates. This suggests that crime is influenced by seasonal or social factors such as festivals, holidays, or weather conditions.

Interpretation:

Crime is not random but follows temporal patterns. Understanding these trends can help law enforcement agencies prepare in advance for high-risk periods.

6.3 Crime Type Analysis

The visualization of crime categories (such as theft, assault, burglary, fraud, etc.) provides insights into the nature of crimes being committed.

- Bar charts clearly indicate that some crime types occur far more frequently than others.
- Property-related crimes generally show higher occurrence compared to violent crimes.

- Less frequent but serious crimes, although lower in number, may have higher social impact.

Interpretation:

Law enforcement efforts should not only focus on the most frequent crimes but also consider the severity and impact of different crime types. Resource allocation should balance frequency and seriousness.

6.4 Location-Wise Crime Distribution

Geographical analysis using maps and location-based charts highlights differences in crime distribution across regions.

- Certain areas consistently report higher crime counts, identifying them as **crime hotspots**.
- Other regions show comparatively low crime rates, indicating safer zones.
- Urban or densely populated areas tend to have higher crime occurrences than rural regions.

Interpretation:

Location plays a crucial role in crime occurrence. Identifying hotspots helps authorities deploy targeted policing, surveillance, and preventive measures in high-risk areas.

6.5 Demographic and Contextual Insights

Where demographic attributes (such as gender, age group, or victim/offender category) are included, the analysis provides additional context:

- Specific demographic groups may appear more frequently associated with certain crime types.
- Repeat patterns suggest social or economic influences behind criminal behavior.

Interpretation:

Crime prevention strategies should consider social awareness programs, education, and community engagement alongside strict enforcement.

6.6 Key Performance Indicators (KPIs) Interpretation

KPI cards in the dashboard summarize critical metrics such as:

- Total number of crimes

- Most common crime type
- Highest crime location
- Year with maximum crime cases

These KPIs provide a quick snapshot of the overall crime situation and help decision-makers grasp key insights instantly.

Interpretation:

KPIs enhance decision-making efficiency by highlighting the most critical crime indicators without requiring detailed analysis of raw data.

6.7 Dashboard Effectiveness and Interactivity

The interactive Power BI dashboard allows users to:

- Filter data by year, location, and crime type
 - Drill down into specific regions or periods
 - Compare multiple dimensions simultaneously
-

6.8 Overall Findings

From the analysis and visual interpretation, the following key findings emerge:

- Crime patterns vary significantly across time and location.
 - Certain crime types dominate the dataset and require focused attention.
 - Visual analytics using Power BI makes complex crime data easier to understand.
 - Data-driven insights can support proactive crime prevention strategies.
-

6.9 Conclusion of Discussion

The discussion and interpretation of results demonstrate that crime data analytics using Power BI is an effective approach for understanding crime behavior and trends. The insights derived from this analysis can assist law enforcement agencies, policymakers, and researchers in making informed decisions, improving public safety, and planning preventive measures.

7. CONCLUSION

7.1 Summary of the Project

This project successfully demonstrated the application of data analytics techniques and visualization tools to analyze crime data using Power BI. The primary objective was to transform raw crime records into meaningful insights that help understand crime patterns, trends, and distributions across time, location, and crime categories. By integrating multiple datasets and applying preprocessing techniques, a comprehensive analytical framework was developed.

7.2 Key Outcomes

The analysis led to several important conclusions:

- Crime occurrences show clear variations across different years and months, indicating that crime follows identifiable temporal patterns.
 - Certain crime types are more prevalent than others, emphasizing the need for focused preventive measures.
 - Geographic analysis revealed crime hotspots, helping identify high-risk areas.
 - Interactive dashboards and KPIs enabled quick understanding of complex crime data.
 - Power BI proved to be an effective tool for handling large datasets and presenting insights in an intuitive and user-friendly manner.
-

7.3 Significance of the Study

The findings of this project highlight the importance of data-driven approaches in crime analysis. By leveraging visualization and analytics, stakeholders such as law enforcement agencies and policymakers can move from reactive responses to proactive crime prevention strategies. The project demonstrates how analytical dashboards can support better decision-making and efficient resource allocation.

7.4 Limitations of the Project

Despite its success, the project has certain limitations:

- The analysis depends on the quality and completeness of the available dataset.
- Some demographic or contextual factors influencing crime may not be fully captured.
- The project focuses on historical data and does not include predictive or real-time crime analysis.

Recognizing these limitations provides direction for improving and expanding the study.

7.5 Future Scope

The project can be further enhanced in several ways:

- Integration of real-time crime data for live monitoring.
 - Application of predictive analytics and machine learning models to forecast crime trends.
 - Inclusion of socio-economic and demographic data for deeper insights.
 - Deployment of the dashboard for public or institutional use with role-based access.
 - Automation of data updates to keep the dashboard current.
-

7.6 Final Conclusion

In conclusion, the Crime Data Analytics project using Power BI effectively illustrates how business intelligence tools can be applied to the field of public safety. The project not only improves understanding of crime data but also showcases the practical value of data analytics in addressing real-world problems. The outcomes of this study reaffirm that visual analytics is a powerful approach for transforming data into actionable intelligence.

8. FUTURE SCOPE

The Crime Data Analytics project using Power BI provides a strong foundation for understanding historical crime patterns and trends. However, there is significant scope to enhance and extend this system in the future to make it more intelligent, scalable, and impactful. The following areas outline the potential future enhancements of this project.

8.1 Integration of Real-Time Data

Currently, the analysis is based on historical crime datasets. In the future, real-time data integration from police databases, emergency call systems, or public safety portals can be implemented. This would allow continuous monitoring of crime activities and enable authorities to respond more quickly to emerging threats.

8.2 Predictive Crime Analytics

Advanced machine learning models can be incorporated to predict future crime occurrences based on historical trends. Techniques such as regression models, time-series forecasting, and classification algorithms can help estimate crime probability in specific locations or time periods, enabling proactive crime prevention.

8.3 Advanced Geospatial Analysis

The project can be enhanced by integrating GIS-based analysis with Power BI maps. Heat maps, buffer zone analysis, and route-based crime tracking can provide deeper spatial insights. This would help identify micro-level crime hotspots and optimize patrol routes.

8.4 Inclusion of Socio-Economic and Demographic Factors

Future versions of the project can include additional datasets such as population density, unemployment rates, education levels, and income distribution. Analyzing crime in relation to these factors can provide a more holistic understanding of the root causes of criminal activities.

8.5 Automation and Data Refresh

Automating data extraction, transformation, and loading (ETL) processes can ensure that the dashboard remains updated without manual intervention. Scheduled data refresh and automated alerts can improve system reliability and usability.

8.6 Role-Based Dashboards

The system can be expanded to provide customized dashboards for different users such as police officers, administrators, policymakers, and researchers. Role-based access control would ensure data security while presenting relevant insights to each user group.

8.7 Integration with Decision Support Systems

The analytics dashboard can be integrated with decision support systems to recommend actions such as resource allocation, patrol planning, and risk alerts. This would transform the project from descriptive analytics to actionable intelligence.

8.8 Public Awareness and Transparency

A simplified version of the dashboard can be made publicly accessible to increase awareness about crime trends. This can help citizens make informed decisions and foster transparency between law enforcement agencies and the public.

8.9 Final Note on Future Scope

In conclusion, the future scope of this project is vast and promising. By incorporating advanced analytics, real-time data, and automation, the system can evolve into a comprehensive crime intelligence platform. Such enhancements would significantly contribute to improved public safety, efficient policing, and data-driven governance.

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10. SOCIAL MEDIA ENGAGEMENT

LINKEDIN POST LINK:

https://www.linkedin.com/posts/sudhirsir_powerbi-dataanalytics-businessintelligence-activity-7405964958523580417-9kGO?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAADPE9-4BlttfGDQHfzfbypBCiQO4Rses19-Q

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"Data tells a story — analysis helps us understand it, and insight empowers us to act."



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11. APPENDICES

This chapter contains supplementary material that supports the main content of the project report. The appendices provide additional details, screenshots, and technical information that enhance understanding but are not included in the main chapters to maintain clarity and readability.

Appendix A: Dataset Description

This appendix provides a brief overview of the crime dataset used in the project.

- **Dataset Name:** Crime Data Analysis Dataset
 - **Source:** NCRB
 - **Format:** Multiple Excel (.xlsx) and CSV files
 - **Time Period Covered:** Multiple years
 - **Key Attributes:**
 - Crime ID
 - Crime Type
 - Date of Occurrence
 - Location / Area
 - City / State
 - Victim / Case Status (if available)
-

Appendix B: Data Preprocessing Steps

This appendix outlines the preprocessing steps applied before analysis:

- Removal of duplicate records
 - Handling missing or null values
 - Standardization of date formats
 - Data type conversion for numeric and categorical fields
 - Merging multiple crime data files into a single dataset
-

Appendix C: Power BI Dashboard Screenshots

This appendix includes screenshots of the interactive dashboards created in Power BI.

- Overall Crime Analysis Dashboard
- Crime Trend Analysis Dashboard

- Location-Based Crime Analysis Dashboard
 - Crime Type Distribution Dashboard
-

Appendix D: Visualizations Used

List of visual elements implemented in Power BI:

- Bar Charts for crime type comparison
 - Line Charts for yearly and monthly trends
 - Pie / Donut Charts for proportion analysis
 - Maps for geographical crime distribution
 - KPI Cards for key metrics
 - Slicers for filtering by year, location, and crime type
-

Appendix E: Measures and Calculations

This appendix lists key measures and calculations used in the project:

- Total Crime Count
 - Crime Count by Year
 - Crime Count by Location
 - Most Frequent Crime Type
 - Year with Maximum Crime Rate
-

Appendix F: System Requirements

- **Software:** Microsoft Power BI Desktop
 - **Operating System:** Windows 10 or above
 - **Hardware Requirements:**
 - Minimum 8 GB RAM
 - Intel i5 Processor or equivalent
 - **Storage:** Minimum 2 GB free space
-

Appendix G: Additional Observations

Any additional findings, assumptions, or constraints encountered during the project are documented here for reference.

Final Note

The appendices serve as a valuable reference section, providing technical depth and visual evidence to support the analysis presented in the main chapters.

12. Project Files & Dashboard Excel file drive

Link: <https://drive.google.com/drive/folders/1z2-uSeJLXDLcA6gFB6jnv7ltl0v17xYhr?usp=sharing>

Dataset Source Link:

<https://www.data.gov.in/ministrydepartment/national-crime-records-bureau-ncrb>